Module 2: Convolution and Filters

1. What is convolution?

Convolution is an operation in which a small matrix called a filter or kernel moves across an image and calculates values from local pixel regions.

At each position:

1. Multiply corresponding image and filter values.
2. Add all the products.
3. Store the result in the output feature map.

The process is repeated across the image.

2. Example of a filter

Suppose an image region contains:

1 2 3

4 5 6

7 8 9

Consider this vertical-edge filter:

-1 0 1

-2 0 2

-1 0 1

Multiply corresponding values and add the products:

1×(-1) + 2×0 + 3×1

+ 4×(-2) + 5×0 + 6×2

+ 7×(-1) + 8×0 + 9×1

The result is:

-1 + 0 + 3 - 8 + 0 + 12 - 7 + 0 + 9 = 8

Therefore, the output value is **8**.

3. How does the filter move?

Image

|

v

Select a small region

|

v

Multiply with filter

|

v

Add all products

|

v

Store one output value

|

v

Move filter and repeat

The collection of output values forms a feature map.

4. What can filters detect?

Filters can learn to respond to:

- Horizontal and vertical edges
- Lines and corners

- Textures
- Curves and shapes
- More complex patterns in deeper layers

The example above uses a manually specified edge filter. In a typical CNN, filter values are learned automatically during training using backpropagation and an optimizer.

5. Why are filters useful?

The same filter weights are reused at different image locations. This is called weight sharing.

It allows a CNN to detect similar patterns in different parts of an image without learning a completely separate filter for every location.

Conclusion

Convolution performs element-wise multiplication followed by summation. By moving filters across an image, CNNs generate feature maps that represent useful visual patterns.